

- Every thing from midterm
 - print, end, f-string
 - input (→ string), int(), float()
 - math operators, math orders
 - Named Constants
 - If, if-else, if-elif-else
 - comparison operators, boolean operators
 - while loop, while True, input validation, break
 - for loop, range(), $x+=1 \rightarrow x=x+1$

Final 45 points (45%)

- ↳ 10 short answer questions (10 point) Easy-med
- ↳ 1 short ^{easy} programming question (5 point)
 - ↳ 2 functions → one specific function
one main function
- ↳ 3 long programming questions (10 point each)
 - ↳ easy, med, hard
 - ↳ 3-5 functions per questions

List

- * list indexing
- * list operators: +, *, in
- * list methods: .append() → add to the end of the list
- * list comprehension: ~~*~~
 - ↳ read & write list comprehension!

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↳ Creating a new list using append (short ver.)

$x = []$
 $\text{for } i \text{ in range}(5):$
 $x.append(i)$

$x = [i \text{ for } i \text{ in range}(5)]$

↳ Can be tuple too
($i \text{ for } i \text{ in range}(5)$)

$x = [0, 1, 2, 3, 4]$

* Useful built-in methods $\rightarrow$ sum(), max(), min(), len()

↳ $\text{print}(\text{sum}([(i+1) \times 2 \text{ for } i \text{ in range}(1, 3)]))$

$\text{sum}([4, 6]) \rightarrow 13 \rightarrow 13$

List comprehension with condition

- add if ... at the end
↳ only append if True.

• $lst = [i \text{ for } i \text{ in range}(5) \text{ if } i \% 2 == 0]$
↳ $[0, 2, 4]$
only even number!

String: "Hello", 'Hi'

↳ an immutable sequence of characters.

• for loop string

• indexing string

(negative index too!)

$x = \text{"Hello"}$

$\text{print}(x[0]) \rightarrow e$

• slicing

• string methods $\rightarrow$

1) .isalpha(), .isalnum(), .isdigit()
.islower(), .isupper()
2) .lower(), .upper()

- String methods →
 - 1) `.islower()`, `.isupper()`
 - 2) `.lower()`, `.upper()`
 - 3) `.strip()`, `.rstrip()`, `.lstrip()`
 - 4) `.split()`

turn string into a list of string

Ex1 `x = "Hello : World : How : are : you?"`
`y = x.split()` ← default split at space
`['Hello', 'World', 'How', 'are', 'you?']`

Ex2 turn "10, 20, 30" → [10, 20, 30]

```

user_input = input("Enter numbers:")
raw = user_input.split(',')
for i in range(len(raw)):
    raw[i] = float(raw[i])
numbers = [float(x) for x in raw]
  
```

Annotations: `raw = input("Enter numbers:").split(',')` → `raw = ['10', '20', '30']`
`raw[i] = float(raw[i])` → `[10, 20.0, 30.0]`

Dictionary

- be able to read
- for loop
- dictionary methods:

- `.keys()` → list of keys
- `.values()` → list of values
- `.items()` → list of a tuple of pair key, value
- `.get(key, default)`

gets(key, default) ^{pair: key, value}

Functions

Create

```
def function_name ( param1, param2, param3 ) :  
    statement  
    statement
```

Optional

store values when you call the function

```
return x, y, z ← optional
```

Send the values out!



Usage:

① return something → need to store in variables

eg. a, b, c = function_name(1, 2, 3)
result = function_name(1, 2, 3)
(x, y, z)

② no return → don't need to store
print_only_function()